

1. The relativistically covariant generalization of the Heisenberg equations are

$$i[P^\mu, \phi(x)] = \partial^\mu \phi(x), \quad (1)$$

where P^μ is the four momentum operator. Using this show that the time translation operator in Hilbert space is given by $T(a) = \exp(iP^\mu a_\mu)$ i.e., $T^{-1}(a)\phi(x)T(a) = \phi(x - a)$

2. In class, until now, we have been dealing only with a real scalar field which describes uncharged scalar particles. In this problem we will consider a complex scalar field which will be seen to describe charged particles. The expansions of the complex scalar fields, $\phi(x)$ and $\phi^\dagger(x)$ in terms of the creation and annihilation operators $a(\mathbf{k})$, $a^\dagger(\mathbf{k})$, $b(\mathbf{k})$, $b^\dagger(\mathbf{k})$ is given by:

$$\begin{aligned} \phi(x) &= \sum_{\mathbf{k}} \frac{1}{\sqrt{2V k^0}} [a(\mathbf{k})e^{-ikx} + b^\dagger(\mathbf{k})e^{ikx}] \\ \phi(x)^\dagger &= \sum_{\mathbf{k}} \frac{1}{\sqrt{2V k^0}} [a^\dagger(\mathbf{k})e^{ikx} + b(\mathbf{k})e^{-ikx}], \end{aligned}$$

where, $a(\mathbf{k})$ and $b(\mathbf{k})$ are associated with the particles (charge +1) and antiparticles (charge -1) respectively.

$\phi(x)$ and $\phi^\dagger(x)$ can be also expressed in terms of two real scalar fields as follows:

$$\phi(x) = \frac{1}{\sqrt{2}}(\phi_1(x) + i\phi_2(x)), \quad \phi^\dagger(x) = \frac{1}{\sqrt{2}}(\phi_1(x) - i\phi_2(x)).$$

You may expand $\phi_l(x)$, $l = 1, 2$ in the usual way as for a real scalar field $l = 1, 2$:

$$\phi_l(x) = \sum_{\mathbf{k}} \frac{1}{\sqrt{2V k^0}} [a_l(\mathbf{k})e^{-ikx} + a_l^\dagger(\mathbf{k})e^{ikx}].$$

Show that

$$a(\mathbf{k}) = \frac{1}{\sqrt{2}}(a_1(\mathbf{k}) + ia_2(\mathbf{k})), \quad b(\mathbf{k}) = \frac{1}{\sqrt{2}}(a_1(\mathbf{k}) - ia_2(\mathbf{k})).$$

Hence, derive the commutation relations for $a(\mathbf{k})$, $a^\dagger(\mathbf{k})$, $b(\mathbf{k})$, $b^\dagger(\mathbf{k})$ from those of $a_l(\mathbf{k})$, $a_l^\dagger(\mathbf{k})$.

3. Charge conjugation for the complex Klein Gordon field is defined by

$$\phi(x) \rightarrow C\phi(x)C^{-1} = \eta_c\phi^\dagger, \quad (2)$$

where C is a unitary operator which leaves the vacuum invariant, $C|0\rangle = |0\rangle$ and η_c is a phase factor. In the notation used in problem 2 for the creation and annihilation operators for the charge +1 and -1 quanta, derive,

$$Ca(\vec{k})C^{-1} = \eta_cb(\vec{k}); Cb(\vec{k})C^{-1} = \eta_c^*a(\vec{k}), \quad (3)$$

and hence show that

$$C|a, \vec{k}\rangle = \eta_c^*|b, \vec{k}\rangle; C|b, \vec{k}\rangle = \eta_c|a, \vec{k}\rangle, \quad (4)$$

where $|a, \vec{k}\rangle$ denotes a state with a single a particle of momentum $\vec{k}$, etc. C is called the charge conjugation operator and it interchanges particles and anti-particles. The phase η_c is arbitrary and is usually set equal to one.